

DD_OSR-02 - Procurement

Version: v1.0 **Bundle:** 08_osr_equipment **Status:** Draft for review **Primary owner:** OSR / procurement operations team
Related DDs: DD_OSR-01, DD_OSR-INSTANCE-01, DD_PLM-CFG-06, DD_EM-02, DD_EM-CFG-03, DD_EM-CFG-04, DD_ICN-01, DD_REP-01, FOUNDATION_DROOLS

MVP simplification baseline: first release may seed opening stock and implement OSR stock/equipment movements before full procurement. Build OSR-02 after stock/instance basics are stable unless launch procurement operations require it.

1. Purpose

OSR-02 defines the SOPHIX procurement process for equipment and stockable materials.

It covers the flow from a stock need to a purchase order, approval, vendor order, delivery receipt, goods receipt into stock, serialized equipment import, discrepancy handling, and reporting.

Use OSR-02 when:

- central warehouse needs more ONTs, routers, STBs, splitters, drop cable, or smart cards;
- procurement creates a purchase order for a vendor;
- finance/procurement approval is needed before sending the order;
- warehouse receives vendor delivery;
- received quantity differs from ordered quantity;
- serialized devices must be created in OSR-INSTANCE-01;
- SKU quantity movements must be recorded in OSR-01.

OSR-02 does not own stock balances. OSR-01 owns SKU/location quantity movement and balance. OSR-INSTANCE-01 owns per-serial equipment records. Procurement is the business process that triggers those owning modules.

2. Scope And Boundaries

Capability	Owner
Procurement request, PO, vendor delivery, receipt workflow	OSR-02
SKU catalog and serialization flag	PLM-CFG-06
Stock movements and balances	OSR-01
Serialized equipment instance creation/lifecycle	OSR-INSTANCE-01
Contractor/vendor/staff reference data	EM-02 / EM business registry
Procurement permissions	EM-CFG-03
Configurable procurement approval	EM-CFG-04
Operator-variable procurement rules	Drools
Notifications to procurement/warehouse teams	ICN-01
Procurement KPIs	REP-01

Boundary rules:

- OSR-02 must not directly update stock_balance.
- OSR-02 records goods receipt, then calls OSR-01 POST /api/stock-movements for quantity stock.
- For serialized SKUs, OSR-02 calls OSR-INSTANCE-01 POST /api/equipment-instances/bulk-import or POST /api/equipment-instances.
- Approval steps are owned by EM-CFG-04. OSR-02 stores only approval references and final outcome.
- Camunda is used only when EM-CFG-04 selected policy mode is BPMN.
- Drools is used for configurable vendor/order/discrepancy policy, not fixed validation.

3. Process Overview

```
flowchart LR
    A["Stock need or manual request"] --> B["Create procurement request"]
    B --> C["Create PO"]
    C --> D["Validate vendor/SKU/budget"]
    D --> E["Evaluate EM-CFG-04 approval"]
    E --> F["PO approved"]
    F --> G["Send PO to vendor"]
    G --> H["Receive vendor delivery"]
    H --> I["Record goods receipt"]
    I --> J["Call OSR-01 stock movement"]
    I --> K["Call OSR-INSTANCE for serialized devices"]
    J --> L["PO closed or partially received"]
    K --> L
```

4. Status Models

4.1 Procurement Order Status

Status	Meaning
DRAFT	PO is being prepared.
VALIDATED	SKU/vendor/basic checks passed.
PENDING_APPROVAL	EM-CFG-04 approval request is open.
APPROVED	PO is approved but not sent.
SENT_TO_VENDOR	PO was sent to vendor.
PARTIALLY_RECEIVED	Some lines received.
RECEIVED	All expected lines received.
CLOSED	PO closed after receipt or business decision.
CANCELLED	PO cancelled before completion.
REJECTED	Approval rejected.

4.2 Goods Receipt Status

Status	Meaning
DRAFT	Receipt capture started.
VALIDATED	Receipt lines passed checks.
POSTED	OSR-01/OSR-INSTANCE writes completed.
DISCREPANCY_PENDING	Quantity/serial/vendor discrepancy requires review.
CANCELLED	Receipt voided before posting.

5. Data Model

5.1 procurement_vendor

What it is for: stores vendor master data used by procurement.

When to use it: create a vendor before raising purchase orders. Vendor records may later be folded into a wider supplier registry, but OSR-02 needs a procurement-facing vendor table for v1.0.

Sample meaning: ven-huawei-ke is an approved equipment vendor for WIK, allowed to supply Huawei ONTs.

Column	Type	Purpose	Sample
vendor_id	TEXT PK	Vendor id.	ven-huawei-ke
operator_code	TEXT	Tenant/operator.	WIK
vendor_code	TEXT	Business code.	HUAWEI_KE
vendor_name	TEXT	Vendor display name.	Huawei Kenya

Column	Type	Purpose	Sample
status	TEXT	DRAFT, ACTIVE, SUSPENDED, RETIRED.	ACTIVE
primary_contact_json	JSONB	Contact details.	{"email": "orders@vendor.example"}
allowed_sku_ids	TEXT[]	Optional SKU allow-list.	["WIK-ONT-HUAWEI-EG8145V5"]
payment_terms_code	TEXT NULL	Commercial payment terms.	NET_30
created_at	TIMESTAMP	Creation timestamp.	2026-06-01T08:00:00Z

Recommended indexes:

- unique (operator_code, vendor_code).
- (operator_code, status).

5.2 procurement_order

What it is for: stores purchase order header.

When to use it: every approved or draft procurement order has one header row. Lines hold SKUs and quantities.

Sample meaning: po-2026-00045 orders 200 Huawei ONTs for WIK central warehouse, requires approval, and will receive into WIK-CENTRAL-NRB.

Column	Type	Purpose	Sample
po_id	TEXT PK	Purchase order id.	po-2026-00045
operator_code	TEXT	Tenant/operator.	WIK
po_number	TEXT	Human-readable PO number.	WIK-PO-2026-00045
vendor_id	TEXT	Vendor id.	ven-huawei-ke
destination_location_id	TEXT	OSR-01 stock location for receipt.	WIK-CENTRAL-NRB
currency_code	TEXT	Currency.	KES
total_estimated_amount	NUMERIC(14,2)	Estimated PO value.	1200000.00
status	TEXT	PO status.	PENDING_APPROVAL
approval_request_id	TEXT NULL	EM-CFG-04 approval request.	apr-po-00045
approval_mode	TEXT NULL	EM-CFG-04 mode.	MULTI_STEP
camunda_process_instance_id	TEXT NULL	BPMN instance if used.	cam-po-00045
requested_by_user_id	TEXT	Requester.	usr-procurement-001
expected_delivery_date	DATE NULL	Expected vendor delivery.	2026-07-15
created_at	TIMESTAMP	Creation timestamp.	2026-06-15T09:00:00Z
approved_at	TIMESTAMP NULL	Approval timestamp.	2026-06-15T11:00:00Z
closed_at	TIMESTAMP NULL	Close timestamp.	null

Recommended indexes:

- unique (operator_code, po_number).
- (operator_code, status, created_at DESC).
- (approval_request_id).

5.3 procurement_order_line

What it is for: stores ordered SKUs, quantities, and unit costs.

When to use it: every SKU in a PO must have one line. This table is the receipt matching reference.

Sample meaning: line po1-001 orders 200 units of Huawei ONT SKU at KES 6000 each.

Column	Type	Purpose	Sample
po_line_id	TEXT PK	PO line id.	po1-001
po_id	TEXT	Parent PO.	po-2026-00045
sku_id	TEXT	PLM-CFG-06 SKU id.	WIK-ONT-HUAWEI-EG8145V5

Column	Type	Purpose	Sample
description	TEXT	Line description.	Huawei EG8145V5 ONT
ordered_qty	INT	Quantity ordered.	200
received_qty	INT	Quantity posted as received.	150
unit_cost	NUMERIC(14,2)	Estimated unit cost.	6000.00
line_amount	NUMERIC(14,2)	Ordered qty times unit cost.	1200000.00
is_serialized	BOOLEAN	Copied from PLM-CFG-06 at order time.	true
status	TEXT	OPEN, PARTIALLY_RECEIVED, RECEIVED, CANCELLED.	PARTIALLY_RECEIVED

Recommended indexes:

- (po_id, status).
- (sku_id).

5.4 goods_receipt

What it is for: stores one vendor delivery receipt transaction.

When to use it: warehouse receives stock against a PO. A PO can have multiple goods receipts.

Sample meaning: gr-001 records the first delivery of 150 ONTs against PO po-2026-00045.

Column	Type	Purpose	Sample
receipt_id	TEXT PK	Goods receipt id.	gr-001
operator_code	TEXT	Tenant/operator.	WIK
po_id	TEXT	Related PO.	po-2026-00045
vendor_delivery_ref	TEXT	Vendor delivery note/ref.	DN-HW-7788
destination_location_id	TEXT	OSR-01 receiving location.	WIK-CENTRAL-NRB
received_by_user_id	TEXT	Warehouse user.	usr-warehouse-001
received_at	TIMESTAMP	Physical receipt time.	2026-07-15T10:20:00Z
status	TEXT	Receipt status.	POSTED
posted_stock_movement_batch_id	TEXT NULL	OSR-01 movement batch ref.	sm-batch-gr-001
created_at	TIMESTAMP	Creation timestamp.	2026-07-15T10:25:00Z
posted_at	TIMESTAMP NULL	Posting timestamp.	2026-07-15T10:27:00Z

Recommended indexes:

- (operator_code, po_id).
- (operator_code, status, received_at DESC).
- unique (operator_code, vendor_delivery_ref).

5.5 goods_receipt_line

What it is for: stores received quantity and serial payload per PO line.

When to use it: each received SKU line has one row. Serialized SKUs include serial/MAC payload for OSR-INSTANCE import.

Sample meaning: gr l-001 receives 150 ONTs and includes 150 serial records.

Column	Type	Purpose	Sample
receipt_line_id	TEXT PK	Receipt line id.	gr l-001
receipt_id	TEXT	Goods receipt id.	gr-001
po_line_id	TEXT	PO line matched.	po l-001
sku_id	TEXT	PLM-CFG-06 SKU id.	WIK-ONT-HUAWEI-EG8145V5
received_qty	INT	Quantity physically received.	150

Column	Type	Purpose	Sample
accepted_qty	INT	Quantity accepted into stock.	148
rejected_qty	INT	Quantity rejected/damaged.	2
serial_payload_json	JSONB NULL	Serial/MAC list for serialized SKUs.	[{"serial": "48575443ABCD1234"}]
quality_status	TEXT	PASS, PARTIAL_FAIL, FAIL.	PARTIAL_FAIL
discrepancy_code	TEXT NULL	Discrepancy reason.	DAMAGED_ON_ARRIVAL

Recommended indexes:

- (receipt_id).
- (po_line_id).
- (sku_id).

5.6 procurement_discrepancy

What it is for: stores receipt/order/vendor discrepancies.

When to use it: received quantity differs from ordered quantity, serials are invalid/duplicate, items are damaged, or vendor delivered wrong SKU.

Sample meaning: pd-001 records 2 damaged ONTs from vendor delivery DN-HW-7788.

Column	Type	Purpose	Sample
discrepancy_id	TEXT PK	Discrepancy id.	pd-001
operator_code	TEXT	Tenant/operator.	WIK
po_id	TEXT	Related PO.	po-2026-00045
receipt_id	TEXT	Related receipt.	gr-001
receipt_line_id	TEXT NULL	Related receipt line.	grl-001
discrepancy_type	TEXT	SHORT_DELIVERY, OVER_DELIVERY, DAMAGED, WRONG_SKU, DUPLICATE_SERIAL, INVALID_SERIAL.	DAMAGED
quantity	INT	Affected quantity.	2
status	TEXT	OPEN, PENDING_APPROVAL, RESOLVED, REJECTED, CLOSED.	OPEN
resolution_action	TEXT NULL	ACCEPT, REJECT_TO_VENDOR, CREDIT_NOTE, REPLACEMENT_PENDING.	REJECT_TO_VENDOR
approval_request_id	TEXT NULL	EM-CFG-04 approval if required.	apr-pd-001
created_at	TIMESTAMP	Creation timestamp.	2026-07-15T10:30:00Z
resolved_at	TIMESTAMP NULL	Resolution timestamp.	null

Recommended indexes:

- (operator_code, status, created_at DESC).
- (po_id).
- (receipt_id).

6. APIs

6.1 Create Purchase Order

```
POST /api/procurement/orders
Idempotency-Key: po-wik-huawei-20260715
Content-Type: application/json

{
  "operatorCode": "WIK",
  "vendorId": "ven-huawei-ke",
  "destinationLocationId": "WIK-CENTRAL-NRB",
```

```

    "currencyCode": "KES",
    "expectedDeliveryDate": "2026-07-15",
    "lines": [
      {
        "skuId": "WIK-ONT-HUAWEI-EG8145V5",
        "orderedQty": 200,
        "unitCost": 6000.00
      }
    ]
  }
}

```

6.2 Submit PO For Approval

POST /api/procurement/orders/{po_id}/submit-approval
 Content-Type: application/json

OSR-02 calls EM-CFG-04:

```

POST /api/approvals/evaluate
Content-Type: application/json

{
  "operatorCode": "WIK",
  "moduleCode": "OSR-02",
  "actionCode": "PROCUREMENT_ORDER_APPROVAL",
  "subjectType": "PROCUREMENT_ORDER",
  "subjectId": "po-2026-00045",
  "context": {
    "totalEstimatedAmount": 1200000.00,
    "currencyCode": "KES",
    "vendorId": "ven-huawei-ke",
    "destinationLocationId": "WIK-CENTRAL-NRB",
    "lineCount": 1
  }
}

```

If approval is required, OSR-02 starts an approval request and waits for callback.

6.3 Approval Outcome Callback

POST /internal/procurement/orders/{po_id}/approval-outcome
 Content-Type: application/json

```

{
  "approvalRequestId": "apr-po-00045",
  "outcome": "APPROVED",
  "decisionComment": "Approved for Nairobi stock replenishment."
}

```

6.4 Record Goods Receipt

POST /api/procurement/goods-receipts
 Idempotency-Key: gr-dn-hw-7788
 Content-Type: application/json

```

{
  "operatorCode": "WIK",
  "poId": "po-2026-00045",
  "vendorDeliveryRef": "DN-HW-7788",
  "destinationLocationId": "WIK-CENTRAL-NRB",
  "receivedAt": "2026-07-15T10:20:00Z",
  "lines": [
    {
      "poLineId": "pol-001",
      "skuId": "WIK-ONT-HUAWEI-EG8145V5",
      "receivedQty": 150,
      "acceptedQty": 148,
      "rejectedQty": 2,
      "serialPayload": [
        {"serialNumber": "48575443ABCD1234", "macAddress": "00:11:22:33:44:55"}
      ],
      "discrepancyCode": "DAMAGED_ON_ARRIVAL"
    }
  ]
}

```

Posting behavior:

- create goods_receipt and goods_receipt_line;
- create discrepancy row if received/accepted/rejected quantities do not match expectations;

- call OSR-01 POST /api/stock-movements for accepted quantity;
- call OSR-INSTANCE-01 bulk import for serialized accepted units;
- update PO line received quantity and PO status.

6.5 Resolve Procurement Discrepancy

POST /api/procurement/discrepancies/{discrepancy_id}/resolve
Content-Type: application/json

```
{
  "resolutionAction": "REJECT_TO_VENDOR",
  "comment": "Two ONTs physically damaged on arrival.",
  "requestedByUserId": "usr-warehouse-001"
}
```

Resolution may call EM-CFG-04 depending on discrepancy value and action.

7. Business Rules

Rule	Description	Reason
R-OSR-02-01	PO vendor must be active.	Prevent ordering from blocked vendors.
R-OSR-02-02	PO SKU must exist in PLM-CFG-06 and be active for operator.	Prevent invalid stock.
R-OSR-02-03	Destination location must exist in OSR-01 and accept vendor receipts.	Avoid wrong stock location.
R-OSR-02-04	PO approval must be evaluated through EM-CFG-04.	Governance.
R-OSR-02-05	Camunda is used only when EM-CFG-04 policy mode is BPMN.	Operational simplicity.
R-OSR-02-06	Goods receipt can post only against approved/sent PO.	Prevent unauthorized stock injection.
R-OSR-02-07	Accepted quantity must create OSR-01 stock movement.	Stock balance integrity.
R-OSR-02-08	Serialized accepted units must create OSR-INSTANCE records.	Serial traceability.
R-OSR-02-09	Duplicate vendor delivery ref must be rejected or return existing receipt.	Idempotency.
R-OSR-02-10	Discrepancy resolution may require approval based on value/action.	Financial and inventory control.

8. Rule Engine Guidance

Use Drools for configurable procurement decisions:

Rule file	Purpose
procurement-order-approval.drl	Decide approval policy based on amount, vendor, SKU family, requester, and destination.
goods-receipt-discrepancy.drl	Decide whether discrepancy is tolerable, blocked, or approval-required.
vendor-risk-policy.drl	Apply operator-specific vendor risk controls.

Example:

```
package com.sophix.osr.procurement

rule "High value procurement requires finance approval"
when
  $po : ProcurementOrderFact(totalAmount >= 1000000.00, currencyCode == "KES")
  $decision : ProcurementDecision()
then
  $decision.setApprovalRequired(true);
  $decision.setApprovalPolicyCode("PROCUREMENT_HIGH_VALUE_FINANCE");
  $decision.addReason("HIGH_VALUE_PO");
end
```

Fixed checks such as required fields, FK existence, quantity math, and idempotency stay in application code.

9. Events

Event	Topic	When emitted
ProcurementOrderCreated	sophix.osr.procurement.order-created	PO created.
ProcurementOrderApprovalRequired	sophix.osr.procurement.approval-required	Approval request started.
ProcurementOrderApproved	sophix.osr.procurement.order-approved	Approval completed.
GoodsReceiptPosted	sophix.osr.procurement.goods-receipt-posted	Receipt posted to OSR-01/OSR-INSTANCE.
ProcurementDiscrepancyOpened	sophix.osr.procurement.discrepancy-opened	Discrepancy created.
ProcurementDiscrepancyResolved	sophix.osr.procurement.discrepancy-resolved	Discrepancy closed.

Use transactional outbox.

10. Frontend/API Consumers

Area	APIs/modules used
Backoffice procurement	OSR-02 order, approval, receipt, discrepancy APIs.
Approval inbox	EM-CFG-04 task APIs.
Warehouse receipt screen	OSR-02 goods receipt, PLM-CFG-06 SKU lookup, OSR-INSTANCE serial validation.
Inventory dashboard	OSR-01 stock balances plus OSR-02 incoming PO status.
Reporting	OSR-02 events and REP-01 datasets.

11. Infrastructure Recommendations

Run OSR-02 inside the OSR/equipment API deployment with OSR-01 and OSR-INSTANCE for a medium budget deployment.

Component	Recommendation
API	Add procurement endpoints to OSR API service.
Database	PostgreSQL OSR schema.
Workers	Shared OSR worker handles outbox and receipt posting retries.
Files	Use foundation file storage for uploaded delivery notes or serial CSV/XLSX.
Approval	EM-CFG-04 shared service.
Rules	Drools/KIE for configurable procurement rules.

AWS notes: use ECS/EKS OSR service, RDS PostgreSQL, S3 for vendor documents/serial uploads, MSK/Kafka for events, Secrets Manager for vendor/API credentials.

12. Acceptance Criteria

- PO lifecycle APIs exist and are idempotent.
- PO approval uses EM-CFG-04; OSR-02 does not hardcode approver steps.
- Goods receipt posts accepted quantity through OSR-01 stock movements.
- Serialized accepted units are created through OSR-INSTANCE-01.
- Receipt discrepancies are tracked and resolvable with approval where configured.

- Drools rules exist for configurable approval and discrepancy decisions.
- Backoffice can create PO, submit approval, receive goods, and resolve discrepancy.